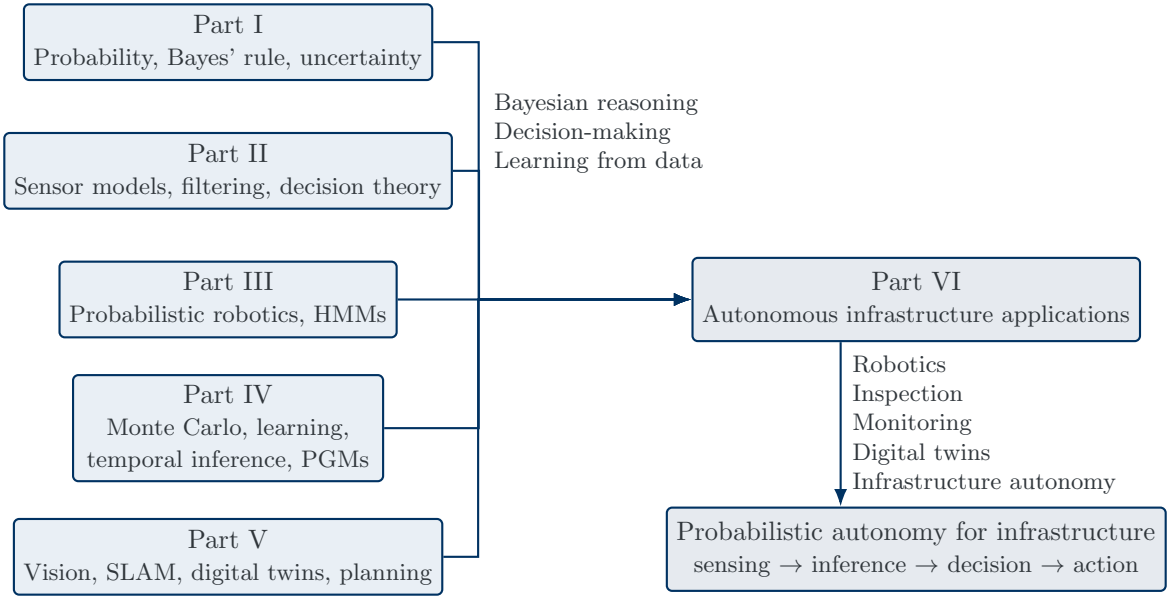
Part I
Probability, Bayes' rule, uncertainty
Part II
Sensor models, filtering, decision theory
Part III
Probabilistic robotics, HMMs
Part IV
Monte Carlo, learning, temporal inference, PGMs
Part V
Vision, SLAM, digital twins, planning
Bayesian reasoning
Decision-making
Learning from data
Part VI
Autonomous infrastructure applications
Robotics
Inspection
Monitoring
Digital twins
Infrastructure autonomy
Probabilistic autonomy for infrastructure
sensing → inference → decision → action